SG No. 223 O_{h}^{3} $Pm\bar{3}n$ [cubic]

* plus set: +[0, 0, 0]

Table 1: Wyckoff site: `2a`, site symmetry: `m-3.`

No.	position	mapping
1	[0, 0, 0]	`[1,2,3,4,5,6,7,8,9,10,11,12,25,26,27,28,29,30,31,32,33,34,35,36]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[13,14,15,16,17,18,19,20,21,22,23,24,37,38,39,40,41,42,43,44,45,46,47,48]`

Table 2: Wyckoff site: `6b`, site symmetry: `mmm..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[1,2,3,4,25,26,27,28]`
2	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,7,8,29,30,31,32]`
3	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[9,10,11,12,33,34,35,36]`
4	$\left[0, \frac{1}{2}, 0\right]$	`[13,14,15,16,37,38,39,40]`
5	$\left[\frac{1}{2}, 0, 0\right]$	`[17,18,19,20,41,42,43,44]`
6	$\left[0, 0, \frac{1}{2}\right]$	`[21,22,23,24,45,46,47,48]`

Table 3: Wyckoff site: `6c`, site symmetry: `-4m.2`

No.	position	mapping
1	$\left[\frac{1}{4}, 0, \frac{1}{2}\right]$	`[1,4,18,19,26,27,41,44]`
2	$\left[\frac{3}{4}, 0, \frac{1}{2}\right]$	`[2,3,17,20,25,28,42,43]`
3	$\left[\frac{1}{2}, \frac{1}{4}, 0\right]$	`[5,8,14,15,30,31,37,40]`
4	$\left[\frac{1}{2}, \frac{3}{4}, 0\right]$	`[6,7,13,16,29,32,38,39]`
5	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[9,12,21,24,34,35,46,47]`
6	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[10,11,22,23,33,36,45,48]`

Table 4: Wyckoff site: `6d`, site symmetry: `-4m.2`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{2}, 0\right]$	`[1,4,18,19,26,27,41,44]`
2	$\left[\frac{3}{4}, \frac{1}{2}, 0\right]$	`[2,3,17,20,25,28,42,43]`
3	$\left[0, \frac{1}{4}, \frac{1}{2}\right]$	`[5,8,14,15,30,31,37,40]`
4	$\left[0, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,13,16,29,32,38,39]`
5	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[9,12,21,24,34,35,46,47]`
6	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[10,11,22,23,33,36,45,48]`

Table 5: Wyckoff site: `8e`, site symmetry: `.32`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,5,9,14,19,24]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[2,7,12,13,17,21]`
3	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[3,8,10,15,20,22]`

continued ...

Table 5

No.	position	mapping
4	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[4,6,11,16,18,23]
5	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	[25,29,33,38,43,48]
6	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	[26,31,36,37,41,45]
7	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	[27,32,34,39,44,46]
8	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	[28,30,35,40,42,47]

Table 6: Wyckoff site: 12f, site symmetry: mm2..

No.	position	mapping
1	$[x, 0, 0]$	[1,4,26,27]
2	$[-x, 0, 0]$	[2,3,25,28]
3	$[0, x, 0]$	[5,8,30,31]
4	$[0, -x, 0]$	[6,7,29,32]
5	$[0, 0, x]$	[9,12,34,35]
6	$[0, 0, -x]$	[10,11,33,36]
7	$\left[\frac{1}{2}, x+\frac{1}{2}, \frac{1}{2}\right]$	[13,16,38,39]
8	$\left[\frac{1}{2}, \frac{1}{2}-x, \frac{1}{2}\right]$	[14,15,37,40]
9	$\left[x+\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	[17,20,42,43]
10	$\left[\frac{1}{2}-x, \frac{1}{2}, \frac{1}{2}\right]$	[18,19,41,44]
11	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}-x\right]$	[21,24,46,47]
12	$\left[\frac{1}{2}, \frac{1}{2}, x+\frac{1}{2}\right]$	[22,23,45,48]

Table 7: Wyckoff site: 12g, site symmetry: mm2..

No.	position	mapping
1	$\left[x, 0, \frac{1}{2}\right]$	[1,4,26,27]
2	$\left[-x, 0, \frac{1}{2}\right]$	[2,3,25,28]
3	$\left[\frac{1}{2}, x, 0\right]$	[5,8,30,31]
4	$\left[\frac{1}{2}, -x, 0\right]$	[6,7,29,32]
5	$\left[0, \frac{1}{2}, x\right]$	[9,12,34,35]
6	$\left[0, \frac{1}{2}, -x\right]$	[10,11,33,36]
7	$\left[\frac{1}{2}, x+\frac{1}{2}, 0\right]$	[13,16,38,39]
8	$\left[\frac{1}{2}, \frac{1}{2}-x, 0\right]$	[14,15,37,40]
9	$\left[x+\frac{1}{2}, 0, \frac{1}{2}\right]$	[17,20,42,43]
10	$\left[\frac{1}{2}-x, 0, \frac{1}{2}\right]$	[18,19,41,44]
11	$\left[0, \frac{1}{2}, \frac{1}{2}-x\right]$	[21,24,46,47]
12	$\left[0, \frac{1}{2}, x+\frac{1}{2}\right]$	[22,23,45,48]

Table 8: Wyckoff site: 12h, site symmetry: mm2..

No.	position	mapping
1	$\left[x, \frac{1}{2}, 0\right]$	[1,4,26,27]
2	$\left[-x, \frac{1}{2}, 0\right]$	[2,3,25,28]

continued ...

Table 8

No.	position	mapping
3	$\left[0,\, x,\, \frac{1}{2}\right]$	[5,8,30,31]
4	$\left[0,\, -x,\, \frac{1}{2}\right]$	[6,7,29,32]
5	$\left[\frac{1}{2},\, 0,\, x\right]$	[9,12,34,35]
6	$\left[\frac{1}{2},\, 0,\, -x\right]$	[10,11,33,36]
7	$\left[0,\, x+\frac{1}{2},\, \frac{1}{2}\right]$	[13,16,38,39]
8	$\left[0,\, \frac{1}{2}-x,\, \frac{1}{2}\right]$	[14,15,37,40]
9	$\left[x+\frac{1}{2},\, \frac{1}{2},\, 0\right]$	[17,20,42,43]
10	$\left[\frac{1}{2}-x,\, \frac{1}{2},\, 0\right]$	[18,19,41,44]
11	$\left[\frac{1}{2},\, 0,\, \frac{1}{2}-x\right]$	[21,24,46,47]
12	$\left[\frac{1}{2},\, 0,\, x+\frac{1}{2}\right]$	[22,23,45,48]

Table 9: Wyckoff site: 16i, site symmetry: .3.

No.	position	mapping
1	$[x,\, x,\, x]$	[1,5,9]
2	$[-x,\, -x,\, x]$	[2,7,12]
3	$[-x,\, x,\, -x]$	[3,8,10]
4	$[x,\, -x,\, -x]$	[4,6,11]
5	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[13,17,21]
6	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, \frac{1}{2}-x\right]$	[14,19,24]
7	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, x+\frac{1}{2}\right]$	[15,20,22]
8	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	[16,18,23]
9	$[-x,\, -x,\, -x]$	[25,29,33]
10	$[x,\, x,\, -x]$	[26,31,36]
11	$[x,\, -x,\, x]$	[27,32,34]
12	$[-x,\, x,\, x]$	[28,30,35]
13	$\left[\frac{1}{2}-x,\, \frac{1}{2}-x,\, x+\frac{1}{2}\right]$	[37,41,45]
14	$\left[x+\frac{1}{2},\, x+\frac{1}{2},\, x+\frac{1}{2}\right]$	[38,43,48]
15	$\left[\frac{1}{2}-x,\, x+\frac{1}{2},\, \frac{1}{2}-x\right]$	[39,44,46]
16	$\left[x+\frac{1}{2},\, \frac{1}{2}-x,\, \frac{1}{2}-x\right]$	[40,42,47]

Table 10: Wyckoff site: 24j, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{4},\, y,\, y+\frac{1}{2}\right]$	[1,18]
2	$\left[\frac{3}{4},\, -y,\, y+\frac{1}{2}\right]$	[2,20]
3	$\left[\frac{3}{4},\, y,\, \frac{1}{2}-y\right]$	[3,17]
4	$\left[\frac{1}{4},\, -y,\, \frac{1}{2}-y\right]$	[4,19]
5	$\left[y+\frac{1}{2},\, \frac{1}{4},\, y\right]$	[5,15]
6	$\left[y+\frac{1}{2},\, \frac{3}{4},\, -y\right]$	[6,13]
7	$\left[\frac{1}{2}-y,\, \frac{3}{4},\, y\right]$	[7,16]
8	$\left[\frac{1}{2}-y,\, \frac{1}{4},\, -y\right]$	[8,14]
9	$\left[y,\, y+\frac{1}{2},\, \frac{1}{4}\right]$	[9,21]
10	$\left[-y,\, y+\frac{1}{2},\, \frac{3}{4}\right]$	[10,23]

continued ...

Table 10

No.	position	mapping
11	$\left[y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[11,22]
12	$\left[-y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[12,24]
13	$\left[\frac{3}{4},\ -y,\ \frac{1}{2}-y\right]$	[25,42]
14	$\left[\frac{1}{4},\ y,\ \frac{1}{2}-y\right]$	[26,44]
15	$\left[\frac{1}{4},\ -y,\ y+\frac{1}{2}\right]$	[27,41]
16	$\left[\frac{3}{4},\ y,\ y+\frac{1}{2}\right]$	[28,43]
17	$\left[\frac{1}{2}-y,\ \frac{3}{4},\ -y\right]$	[29,39]
18	$\left[\frac{1}{2}-y,\ \frac{1}{4},\ y\right]$	[30,37]
19	$\left[y+\frac{1}{2},\ \frac{1}{4},\ -y\right]$	[31,40]
20	$\left[y+\frac{1}{2},\ \frac{3}{4},\ y\right]$	[32,38]
21	$\left[-y,\ \frac{1}{2}-y,\ \frac{3}{4}\right]$	[33,45]
22	$\left[y,\ \frac{1}{2}-y,\ \frac{1}{4}\right]$	[34,47]
23	$\left[-y,\ y+\frac{1}{2},\ \frac{1}{4}\right]$	[35,46]
24	$\left[y,\ y+\frac{1}{2},\ \frac{3}{4}\right]$	[36,48]

Table 11: Wyckoff site: 24k, site symmetry: m..

No.	position	mapping
1	$[0,\ y,\ z]$	[1,28]
2	$[0,\ -y,\ z]$	[2,27]
3	$[0,\ y,\ -z]$	[3,26]
4	$[0,\ -y,\ -z]$	[4,25]
5	$[z,\ 0,\ y]$	[5,32]
6	$[z,\ 0,\ -y]$	[6,31]
7	$[-z,\ 0,\ y]$	[7,30]
8	$[-z,\ 0,\ -y]$	[8,29]
9	$[y,\ z,\ 0]$	[9,36]
10	$[-y,\ z,\ 0]$	[10,35]
11	$[y,\ -z,\ 0]$	[11,34]
12	$[-y,\ -z,\ 0]$	[12,33]
13	$\left[y+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[13,40]
14	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[14,39]
15	$\left[y+\frac{1}{2},\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,38]
16	$\left[\frac{1}{2}-y,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[16,37]
17	$\left[\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	[17,44]
18	$\left[\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	[18,43]
19	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	[19,42]
20	$\left[\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	[20,41]
21	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[21,48]
22	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[22,47]
23	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[23,46]
24	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[24,45]

Table 12: Wyckoff site: `48l`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$[-x,\ -y,\ z]$	`[2]`
3	$[-x,\ y,\ -z]$	`[3]`
4	$[x,\ -y,\ -z]$	`[4]`
5	$[z,\ x,\ y]$	`[5]`
6	$[z,\ -x,\ -y]$	`[6]`
7	$[-z,\ -x,\ y]$	`[7]`
8	$[-z,\ x,\ -y]$	`[8]`
9	$[y,\ z,\ x]$	`[9]`
10	$[-y,\ z,\ -x]$	`[10]`
11	$[y,\ -z,\ -x]$	`[11]`
12	$[-y,\ -z,\ x]$	`[12]`
13	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[13]`
14	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[14]`
15	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[15]`
16	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[16]`
17	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[17]`
18	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[18]`
19	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[19]`
20	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[20]`
21	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[21]`
22	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	`[22]`
23	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	`[23]`
24	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	`[24]`
25	$[-x,\ -y,\ -z]$	`[25]`
26	$[x,\ y,\ -z]$	`[26]`
27	$[x,\ -y,\ z]$	`[27]`
28	$[-x,\ y,\ z]$	`[28]`
29	$[-z,\ -x,\ -y]$	`[29]`
30	$[-z,\ x,\ y]$	`[30]`
31	$[z,\ x,\ -y]$	`[31]`
32	$[z,\ -x,\ y]$	`[32]`
33	$[-y,\ -z,\ -x]$	`[33]`
34	$[y,\ -z,\ x]$	`[34]`
35	$[-y,\ z,\ x]$	`[35]`
36	$[y,\ z,\ -x]$	`[36]`
37	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[37]`
38	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[38]`
39	$\left[\frac{1}{2}-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[39]`
40	$\left[y+\frac{1}{2},\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	`[40]`
41	$\left[\frac{1}{2}-x,\ \frac{1}{2}-z,\ y+\frac{1}{2}\right]$	`[41]`
42	$\left[x+\frac{1}{2},\ \frac{1}{2}-z,\ \frac{1}{2}-y\right]$	`[42]`
43	$\left[x+\frac{1}{2},\ z+\frac{1}{2},\ y+\frac{1}{2}\right]$	`[43]`
44	$\left[\frac{1}{2}-x,\ z+\frac{1}{2},\ \frac{1}{2}-y\right]$	`[44]`
45	$\left[\frac{1}{2}-z,\ \frac{1}{2}-y,\ x+\frac{1}{2}\right]$	`[45]`
46	$\left[\frac{1}{2}-z,\ y+\frac{1}{2},\ \frac{1}{2}-x\right]$	`[46]`

continued ...

Table 12

No.	position	mapping
47	$\left[z+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-x\right]$	[47]
48	$\left[z+\frac{1}{2},\ y+\frac{1}{2},\ x+\frac{1}{2}\right]$	[48]